

Aadit Sengupta

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EDUCATION

University of Michigan

Bachelor of Science in Computer Science, and Psychology | GPA: 3.61/4

Coursework: Data Structures and Algorithms, Linear Algebra, Software Engineering, Computer Org, Machine Learning

Ann Arbor, MI

Aug. 2024 – May 2028

EXPERIENCE

AI Software Engineer and Research Intern

April 2026 – August 2026

Lexsi.ai — Python, Java, TensorFlow, GitHub

London, England

- Built ICS, a training-free, parameter-free probe detecting LLM compliance from residual-stream activations, matching or beating tuned guard models (Llama Guard 3, Qwen3Guard) across 20 regulatory domains at 1000x lower calibration cost.
- Applied ICS across 6 finance/legal regulatory domains (cross-border payments, GDPR, HIPAA, EU AI Act), critically evaluating model outputs as part of a 20-domain audit that achieved 0.952 mean AUROC with zero trained parameters
- Co-authored a research paper based on these findings that has been selected for publication in EMNLP 2027.

AI Research Intern

April 2025 – August 2025

Arya.ai — Python, Java, TensorFlow, GitHub

London, England

- Implemented AI interpretability techniques into a previously black-box system, demonstrating a 50% increase in model effectiveness and enabling more transparent and data-driven decision-making.
- Co-Authored a research paper (Interpretability as Alignment) on 4 major alignment approaches (mechanistic interpretability, RLHF, red teaming, constitutional AI) synthesizing key findings across academic and industry research to evaluate their practical use cases, effectiveness, and potential for scalable alignment.
- Selected by **EurIPS** (AI conference in Europe) and research paper presented in their 2025 Copenhagen conference.

AR/VR Software Development Intern

April 2024 – August 2024

Mesmerise VR — C++, TypeScript, Github

London, England

- Collaborated with and led a six-person development team using Agile to build a card game in Unity for Oculus Quest 2 in C++, with augmented reality via pass-through interactivity, and spatial mapping to enhance immersion.
- Collected and analyzed user feedback showing that 85% of participants demonstrated greater retention of Oculus Quest 2 development concepts when using the interactive training compared to traditional learning methods.

PUBLICATIONS

What Do Compliance Detectors Read?

EMNLP 2027

Introduced ICS, a training-free activation probe for LLM regulatory compliance

2026

Interpretability as Alignment: Making Internal Understanding a Principle

DOI

Position Paper on the importance of AI Alignment on Black Box model interpretability

Aug. 2025

Time Dilation & Rotation Around Black Holes

DOI 1 | DOI 2

Two papers applying Relativity: rotation/dilatation around a Kerr or Schwarzschild black hole

Aug. 2025/26

PROJECTS

Stock Movement Predictor | Python, scikit-learn, pandas, yfinance, **GitHub**

August 2026

- Built a data pipeline pulling market data via yfinance and engineered RSI, MACD, and Bollinger Band features to predict stock direction using Random Forest and Gradient Boosting classifiers
- Used time-series cross-validation to avoid lookahead bias and show that engineered features gave no edge over a naive baseline, a result consistent with weak-form market efficiency

RAG Research Assistant | Python, LangChain, ChromaDB, Streamlit, **GitHub**

June 2026

- Built a Retrieval-Augmented Generation system orchestrating document embeddings via ChromaDB and LLM calls via the Groq API to answer questions over my research paper knowledge base
- Diagnosed a PDF extraction bug hurting retrieval accuracy through iterative debugging, then deployed a Streamlit interface enabling real-time querying of the underlying document corpus

Oppia Open Source Contribution

April 2026 – May 2026

- Diagnosed and merged a bug fix into Oppia, an open-source education platform serving 400k+ learners

AI & Recidivism Data Analysis | Python, pandas, NumPy, Matplotlib, **GitHub**

June 2024 – March 2025

- Conducted data preprocessing and integration from multiple sources to analyze AI-assisted recidivism predictions
- Evaluated AI model outputs to research on the social and ethical implications of predictive systems

TECHNICAL SKILLS

Languages: C/C++, Java, Python, TypeScript, SQL, Kotlin, HTML/CSS, C#, JavaScript, Rust, R, Swift

Developer Tools: Git, GitHub, MySQL, MySQLWorkbench, .net, Angular, Anaconda, AWS, VS Code, Visual Studio, PyCharm, Outlook, IntelliJ, JSON, GraphQL, Eclipse, Unity, Fusion360, Machine Learning, Trello, Excel, Linux

Libraries/ Frameworks: TensorFlow, PyTorch, Node.js, React, pandas, Agile (Scrum), NumPy, Matplotlib

Skills: Natural Language Processing, AI Research, Mechanistic Interpretability, Retrieval-Augmented Generation, AI Safety & Regulatory Compliance, Machine Learning, Time-Series Analysis, AR/VR Development